What a Dark-Energy-Tied Galaxy Law Demands of Its Cold Sector: No-Go Theorems, a Covariant Parent, and the Labelled-Domain Construction

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Abstract

We report one night of structural lanes on a gravity framework whose galactic law is $g_N = g^2/(g + a_0)$ with $a_0 = \frac{1}{2}c\sqrt{G\rho_{\rm DE}}$ (the $\frac{1}{2}$ is fitted). Its extra gravitating mass is a collisionless "cold energy" medium, not a particle. The question was what any consistent theory of that medium must look like. Every lane froze its criteria before computing and ran a mutation control. Post-hoc amendments are disclosed, and both a_0 footings (9.36×10^{-11} and $1.13 \times 10^{-10}\,\mathrm{m\,s^{-2}}$) are reported and never pooled. The companion note PAPER56 has the data-side results of the previous day.

1. **A covariant parent.** A fluid effective theory of comoving labels is a covariant parent of the record's static field equation. It reproduces the law to 2.2×10^{-16} and has an isotropic dust background, a derived field-aligned stress, linear well-posedness and slip $\leq 1.54 \times 10^{-5}$, under five conditions (CONDITIONAL). The medium's defining identity holds exactly through shell crossing, so the record's postulate that freed cold energy stays free cannot be derived from it.
2. **Four no-go results.** (D-T) If the medium's excess is the law's phantom, the law's deep limit forces the tether to cancel the medium's own gravity: no linear self-gravity, and no gravity between tethered components. (G) No local Lagrangian in the medium's expansion and tether invariant is ghost-free, gives the law and stays off in the linear universe, unless every static region carries a vacuum offset 15–25 times the slip bound. (L-G, T-O2) A local second-order field that supplies the deep law must carry an external-field effect, so with no satellite external-field effect the law's extra mass must be the cold energy itself. (S) Any tether without membership labels makes a satellite's medium hover while its stars fall, which the satellite data exclude.
3. **Non-local parents.** No causal non-local parent has the record's cosmological engine as its exact limit. The engine's source relocation cannot be conserved, its RMIN cut switches the law off for 128 of 175 SPARC galaxies, and its phantom is a global QUMOND field with an external-field effect. On the positive side, a parent that changes the source of gravity has no hover, a^1 growth and a forced zero-sum.
4. **Energy.** The record's settling does not conserve energy (CFG932). No energy-conserving repair that keeps the released settling energy inside matter brings an infalling halo onto the law (CFG935: NO REPAIR under the frozen rule). Read post hoc, the only form found that holds the law needs a dynamical dark-energy partner plus a thermal reservoir.
5. **The labelled-domain construction, tested.** Only labelled domains, one tether per collapsed object, pass the satellite test. Written down (CFG950), they remove the satellite hover and the linear-growth flag, but energy at label events closes only through a non-dynamical label energy: NOT CONSISTENT AS SPECIFIED under the frozen rule, CONDITIONAL only under a post-hoc amendment. In clusters they fail (CFG951): members' labelled media source the host's law, and the predicted eRASS1 outer shear overshoots ($Z = -9.3$ canonical, -11.9 alternative). The engine design is SMOKE PARTIAL (CFG952): the momentum-conserving draw is infeasible, and at 1024^3 the engine applies the law only above $2.19 \times 10^{11}\,M_\odot$ of retained baryons. No complete parent survives, and the two-footing shear pass of the earlier notes remains a computational result of an engine whose physical parent is not established.
6. **Data.** Faint gas-rich rotators do not reach the dwarfs' accelerations (not diagnostic). Coma ultra-diffuse galaxies sit above the framework's derived ceiling, but not at 3σ once systematics are included. Clusters modelled as untethered cold energy match the shear stack at the primary corner, but the match is fragile.

1 Setting

- **The field (S2).** The record's static field equation is $\mathcal{L} = -|\nabla\Phi - K\mathbf{P}|^2/(2K) - \rho_b\Phi - W_{\rm rem}(u) - (\gamma/2)|\nabla\mathbf{P}|^2$, with $K = 4\pi G$, $u = K|\mathbf{P}|$, $KW'_{\rm rem}(u) = u^2/(a_0 - u)$ and $\gamma = KL_*^2/2$. Its rest state is the law, which is the simple interpolating function of Famaey & Binney (2005).
- **The medium (S3).** Cold energy is a collisionless rest-mass medium whose displacement is $\mathbf{P}$, with $\partial_t\mathbf{P} = -\mathbf{j}$ and $\nabla\cdot\mathbf{P} = \rho - \bar\rho$. So its excess density is the law's phantom. Its amount, $\Omega_c/\Omega_b = 5.364$, is measured, not derived.
- **The postulate (S4).** Cold energy whose binding becomes positive is permanently free. The record's working rule H-EX makes the cold share of expelled baryons free. Below, "tether" means the force that holds the medium on the law, and an external-field effect (EFE; Milgrom 1983) is a change of a system's internal dynamics caused by a uniform external field.

- **The engine (class A).** The structure behind the shear passes of PAPER55 and PAPER56 comes from a separate particle-mesh engine. Every particle is kicked by one potential. The law switches on in turnaround catchments around resolved peaks. A phantom source is computed from the retained baryons, and a zero-sum source, filled by an instantaneous shell draw, supplies the extra mass in each catchment.

The night's lanes asked whether one covariant theory contains both S2+S3 and class A, and, if not, what the cold sector must be.

2 The canonical action and its covariant parent (CFG938)

CFG938 wrote the medium as a relativistic continuum, a fluid effective theory of comoving label fields. The derived action is

$$S = \int \sqrt{-g}\Big[\tfrac{c^4 R}{16\pi G} - mc^2 n - W_{\rm full}(u) - \tfrac{3\gamma}{4K^2}\tau_{\mu\nu}\tau^{\mu\nu} + \lambda\big(D_\mu D^\mu \chi - Kmn\big)\Big] + S_b,$$

with $W_{\rm full}(u) = [-a_0 u - a_0^2 \ln(1 - u/a_0)]/K$, $u = |D\chi|$, $\gamma = KL_*^2/2$, and $\tau_{\mu\nu}$ the traceless medium-frame tidal tensor of χ.

- **Forced.** A dust background ($w = 0$). The cosmic frame as the medium's rest frame. Isotropy ($\mathbf{P} \equiv 0$ on FRW). $\mathbf{P}$ is the electric part of the 2-form potential of the medium's conserved current, so $\nabla\cdot\mathbf{P} = \rho - \bar\rho$ and $\partial_t \mathbf{P}_L = -\mathbf{j}_L$ hold exactly and linear $\mathbf{P} = -\bar\rho\xi$. The tether $\mathbf{P}\cdot\nabla\Phi$ is the medium's ordinary gravitational coupling. Some extra structure is also forced, because $u = K|\mathbf{P}|$ is a non-local, enclosed-mass quantity that no strain function can express. Finally $\gamma > 0$ is forced by well-posedness.
- **Chosen.** That extra structure is a non-propagating potential χ with a multiplier λ; W itself (S2's input; κ fitted); and the stiffness realisation. A local equation-of-state form of the stiffness is excluded by nucleosynthesis if the medium is present then (ratio 1.3×10^3–2.3×10^6 at $z = 4 \times 10^8$); the χ-tidal form has no background.
- **Results.** The law is reproduced to 2.2×10^{-16} on NGC 3198 (43 radii, both footings). The FRW background has $w = 0$ exactly and isotropic stress. The static stress, $p_r = -W$ and $p_t = p_r + Du/K$, is exactly the record's field-aligned stress (CFG660), now derived. The ideal action has no settling drift: the record's drift is an overdamped dissipative extension, and the energy books close with the tether energy plus heat, which CFG932 omitted. The slip is $\le 1.54 \times 10^{-5}$. The dispersion relation $\omega^2 = \gamma\rho_0 k^2 + K\rho_0[R'\cos^2\theta + (R/u)\sin^2\theta] \ge 0$ cures CFG660's $\sqrt{k}$ growth. Inertia is positive iff $\rho_c c^2 > D(u)u/K$, with margin 5.9×10^4–2.5×10^8 on the test systems.
- **Five conditions.** (1) the extra non-propagating fields χ and λ; (2) $\gamma > 0$ and $\rho_c c^2 > D(u)u/K$; (3) the stiffness on χ's tidal tensor; (4) matching to S2 only in the longitudinal sector, which is exact for spherical systems; (5) single-stream flow with $\rho_c > 0$.
- **Three flags of equal weight.** C3 FAIL is inherited from the law. The tethered medium has no linear self-gravity: growth goes as $a^{0.61}$ against a^1 in the engine, corrected to $a^{0.694}$ by CFG941, so the shear pass does not transfer to this parent. And tethered components do not gravitate on each other: in a host's medium field the differential free fall is 0.03–$0.17\,a_0$.
- **Scorecard.** On the record's 20-gate contract for one action (CFG678): PASS A5, C2; CONDITIONAL A1, A2, B3, C1; FAIL C3 (inherited from the law); UNTESTED A3, A4, A6, B1, B2, D1, D2.
- **Particle in disguise.** Yes for the mass: the term $-mc^2 n$ is cold matter. The tether sector has no propagating mode.

Lane class: COVARIANT PARENT CONDITIONAL; all 30 checks pass and the mutation controls bite. Two post-hoc items are disclosed: A1 (an integrator step fix; run 1 kept) and A2 (the tethered–tethered gravity check, added during scoring).

3 The medium's exact kinematics (CFG939) and its energy books (CFG932, CFG935)

CFG939 treated $\mathbf{P}$ as the exact Lagrangian displacement of a collisionless medium. It asked whether S4 is just the multi-stream regime after shell crossing (hypothesis H-SC).

- **The identity is exact.** With $\mathbf{P} = -\int \mathbf{j}\,dt$ (S3's total current), $\nabla\cdot\mathbf{P} = \rho - \bar\rho$ holds exactly through shell crossing: the 1-D multi-stream identity holds to 9×10^{-16}. To first order $\mathbf{P} = -\bar\rho\Psi$, and in spherical symmetry $P_r = \bar\rho(q^3 - r^3)/(3r^2)$.

- **Multi-stream matter stays tethered.** The kinetic term is forced, and every element feels $-\nabla\Phi_b + R(K|\mathbf{P}|)\hat{\mathbf{P}} - \gamma\nabla\rho_c$ in any stream. The law state itself is reached by phase mixing: 100% of the inner mass has crossed, and the profile is at 0.999 of the law at $2r_M$. So the H-SC premise is FALSE.
- **S4 is not derivable.** S4's free population is not a regime of the single medium; it needs a coupling the action does not contain (CFG888's unwritten coupling). The frozen gate C1 needs $m/m_{\rm law}$ within ±0.05 at $5r_M$. The primary run gives 0.949, outside by 0.001. The first run gave 0.966, and the report-only runs give 0.958, 0.956, and 0.994 at $t = 6000$. So S4 is unassigned by the frozen rules (C1 borderline), never derived, and not derivable on the analytic test C3 and on C1 in 4 of 5 runs. C2(a) is 0.43–0.52 against a 0.9 bar. The H-SC mapping also MISALIGNS with the data in all 16 cells.
- **Derived: an unbinding threshold.** Sudden baryon loss unbinds a medium at rest on the law iff $f > f_{\rm crit}(x) = 1 - (1-x)(\mathrm{artanh}\,\sqrt{x} - \sqrt{x})/x^{3/2}$, with $x = u/a_0$. This is 2/3 in the deep regime (0.673–0.691 at dwarf $r_{1/2}$, 0.774 for a Milky-Way-like system). The unbound medium leaves, which gives a deficit, not an excess.
- **Derived: a flat potential.** The law state sits in a flat potential $\mu = -(2/3)GM_b/R_e$ (numerically -0.01339 against -0.01333).
- **Derived: no capture.** A system captures no cold energy iff it moves through the settled medium faster than its own baryonic escape speed. Tidal dwarfs (21–53 km/s) and NGC 1052-DF2/DF4 (22–23 km/s) against $v_{\rm rel} \geq 100$ km/s therefore come out Newtonian.
- **The stiffness is a barrier.** The cold fluid breaks by a gradient catastrophe at finite density, and crossing is suppressed inside $r_{\rm sub} = L_*^2/(4r_M)$. That protects the faintest systems, which is backwards for H-SC.
- **Energy.** The Lagrangian-map dynamics conserves energy, to 1.3×10^{-4} with crossings. Settling runs by phase mixing plus ejection: 44% of the infall mass leaves (21.4 of 48.9 M_b). So the record's energy leak belongs to its class-A drift law, not to S2+S3. A conservative tether only expels mass and never holds a surplus, so the excess family stays unexplained.

Checks: 30/30 symbolic and 11/11 numerical pass; the five mutation teeth bite; two earlier failed runs are kept. A1 was pre-result. A2 is post hoc: the Newtonian control was moved to a quarter step after a failed energy check, and three report-only runs were added.

The energy books of the record's settling (CFG932). The record's settling obeys $d(K+W)/dt = P_{\rm drift} + P_{\rm bath}$: Q1 is NON-CONSERVING. The two-sided drift injects energy when a surplus is pushed out (1-D $\Delta W/|W_0| = +0.593$, identity 1.9×10^{-3}). That energy is booked to a dark-energy partner, which an exact Λ cannot supply, and the baryons never enter the books. At the record's own supply ($f_{\rm ret} = 0.10$) every dwarf's native collapse debris lies below the law's phantom over its stellar body, with a retained excess at r_h of ≤ 0.001 dex. A dwarf excess of the observed size (+0.18 to +0.39 dex) appears only at about 100 times that supply. The shear pass is the $\varepsilon \to 0$ limit of energy-limited settling, not a test of it. The energy-limited hypothesis (H-ELS) is post hoc and so is the lane's amendment A1. An external audit (CFG932-audit) reproduced the energy identity and the class arithmetic. It found an endpoint-accounting defect in the high-supply bracket, so CFG932's ε window is not an audited physical window. Q1 and the record-supply finding are not affected, and the independent CONTRADICTED verdict on H-ELS (CFG930) stands.

An energy-conserving repair (CFG935): NO REPAIR. Does any energy-conserving form of settling keep the law as the settled state? CFG935 used a 1-D shell toy ($G = a_0 = 1$, Plummer baryons, the simple law, 10^4 shells) and the density gate D, the median of $\log_{10}(\bar{\rho}_c/\rho_{\rm ph})$ over $r = 1$–9 (on the law iff $|D| \leq 0.1$).

- **The energy theorem.** An infall start carries more energy than the settled law: +0.165 of $|E_{\rm law}|$ against a Jeans-supported law and +0.532 against the cold S2 ground state. A cold surplus start carries -1.60, a deficit that needs an energy input.
- **Five repairs, all NOT VIABLE.** Each books every source and sink, and the mutation runs show what is lost with the booking off. None that keeps the released energy inside matter brings an infall start onto the law. A Hamiltonian version of the record's two-sided drift (FIX-1, an energy λF) ends at $D = -0.216$, with a reaction on the baryons of $13\,g_{\rm law}$ (125 at $\lambda = 100$). The canonical S2 tether has the law as its exact unique ground state, but infall carries 53% of the binding energy as excess; it is shed by unbinding 41% of the supply, and the warm remnant sits at $D = -0.271$. A one-sided drift with booked heat drives a circulation ($D = -0.102$ from the settled start, -0.355 on infall). A radiation channel runs away into gravothermal collapse and would need a new gapless field, since the record's own field $\mathbf{P}$ is gapped (CFG760–762). A dark-energy partner, as frozen, keeps exact books but ends off the law ($D = -0.145$ / -0.215).
- **Post hoc, no verdict weight.** With the record's stochastic, isotropising bath, the dark-energy partner holds the law from both starts ($D = +0.003$ / $+0.010$), at the price of two new reservoirs: a dynamical dark-energy partner (an exact Λ cannot exchange energy) and a thermal reservoir at the local Jeans temperature. The lane

Table 1: Structural results of the night. "Phantom" is the law's extra mass; "tether" is the force that holds the medium on the law.

result (lane)	statement	rules out
D-T (CFG941)	a cold static medium whose phantom is its own mass, held by an always-on tether: the deep law forces $c_2 = 1$	linear self-gravity and gravity between tethered components in any parent reducing to S2+S3
G (CFG944)	any local $L(\theta, u)$: ghost freedom, the static law and a tether off in the linear universe force $B \geq W(u_*)$	option 1, the collapse-gated tether, for local Lagrangians
L-G, T-O2 (CFG946)	a local second-order field that supplies the deep law has an external-field effect	option 2, cold energy that is not the phantom, for local static selection
S (CFG945)	at leading order every label-free tether hovers	every tether without membership labels
T-M, T-R, T-X (CFG947)	the engine's relocation is not conservable; RMIN cuts the law; its phantom is global	option 3 as an exact parent of the engine

reads this as REPAIR NEEDS NEW PHYSICS.

- **Physical sizes, per footing.** The excess is 0.14–0.40 (canonical) / 0.16–0.39 (alternative) of each catchment's binding energy. The dark-energy exchange needed since $z = 1$ is $\Delta\rho_{\rm DE}/\rho_{\rm DE} = 0.80$–$1.10\times10^{-6}$ / 0.92–1.25×10^{-6}: allowed, and unmeasurable.

So the settled law needs an energy sink outside the matter sector. K0 reproduces CFG932's drift test exactly; all three mutation teeth bite.

4 The theorems

Table 1 lists the five structural results. Each is stated below with its assumptions as written in the lane files.

4.1 Theorem D-T: the tether cancels the medium's own gravity (CFG941)

- **Assumptions.** A cold static medium is held by any always-on tether force $F(u)$, and its phantom is its own gravitating mass, so $g = g_N + u$.
- **Statement.** Equilibrium gives $F(u) = g$. With the law this is $F(u) = a_0 u/(a_0 - u) = u + u^2/a_0 + \dots$ For any law with the deep limit $g_N \to g^2/a_0$, the linear coefficient is 1, whatever the higher terms. Only $c_2 = 1$ gives the baryonic Tully–Fisher relation: with $F = c_2 u + u^2/a_0$ the limit is 1 at $c_2 = 1$, infinite at $c_2 = \frac{1}{2}$ (an acceleration floor) and 0 at $c_2 = 2$ (rescaled Newton). The linear self-gravity coefficient is $(1 - c_2)K\rho_c$, and the mutual term between two tethered components carries the same factor. So the law implies no linear self-gravity of the settling medium and no gravity between tethered components.
- **The diagnosis.** This is not a double count of the source ($\nabla^2\Phi = K(\rho_b + \rho_c)$ appears once). It is a force cancellation: the quadratic term of $W_{\rm full}$ gives the tether a term that cancels the medium's own Newtonian pull.
- **What it rules out.** No repair passes every test. Replacing $W_{\rm full}$ (R1), splitting free and tethered populations (R2: growth needs a tethered fraction $f_t \leq 0.060$, so the free mass, $\times15.6$, enters the law's source), a density gate (R3n) and a range gate (R3ℓ: the required windows lie 4.0–6.7 decades apart) are NOT REPAIRABLE. A turnaround gate (R3θ) is PARTIAL: the law, the background, a^1 growth and slip pass, but it has a ghost at $k_g R = 0.23$–2.19.
- **The hover lemma.** In any static cold tethered equilibrium, medium at rest feels no net force. A satellite's tethered medium therefore feels no host force while its stars do: $\Delta a/g_{\rm host} = 1.02$–$1.11$. The stars leave their medium in 9–106 Myr, and a static satellite would lose 0.10–0.73 dex of phantom, an external-field effect. So the record's reading that satellites show no EFE (CFG925 R3) is not respected. This is an estimate with its assumptions listed.
- **Shear.** The class-A engine is not the non-relativistic limit of any tested local parent, so the T1x/T2x/NC1x shear passes do not transfer. Lane class: STRUCTURAL OBSTRUCTION. Amendment A1 was pre-result (a numerical test point moved to respect the law's own $\sqrt{y}$ correction).

4.2 Theorem G: no local gate cures the ghost (CFG944)

- **The ghost.** Any medium term $F(\theta, u)$ in the expansion θ has the rank-one second variation $(F_{\theta\theta}/2J_0)(\delta\dot{J} - \theta_0\delta J)^2$, so the kinetic coefficient is $\rho_0 + F_{\theta\theta}k^2$. For R3$\theta$ it degenerates, with $\omega^2 \propto 1/(k - k_g)$ unbounded below. On NGC 3198 $k_g r = 5 \times 10^{-4}$–1.2×10^{-2}, so the ghost covers every galactic scale.
- **Assumptions.** $L(\theta, u)$ is any C^2 local function. (d) Ghost freedom at all k, $L_{\theta\theta} \geq 0$. (a) $L(0, u) = -W(u) + L(0, 0)$ on static states. (c-amp) $L(\theta, u) = L(\theta, 0)$ near a cosmic θ_T where linear states with $u = u_*$ occur. (c) $L(\cdot, 0)$ linear on the cosmic range, with T that linear function.
- **Statement.** $B \equiv L(0, 0) - T(0) \geq W(u_*)$. The proof is one line of convexity. Every static region then carries a vacuum term $T_{\mu\nu} = Bg_{\mu\nu}$. On the canonical footing $u_* = 0.774\, a_0$ and $B \geq 0.712\, a_0^2/K$, against the slip-allowed 0.0476 (15.0 times). On the alternative footing $B \geq 0.624\, a_0^2/K$ against 0.0247 (25.3 times).
- **What it rules out.** Every cure is NOT CURABLE. An aether term needs $c_\theta \leq 8.9 \times 10^{14}$ kg/m for growth, while the ghost bound is $\geq 1.6 \times 10^{24}$. Density, χ and latch gates fail, and so does every gate profile (a concavity theorem). The only globally ghost-free form (C4) switches the tether on in the linear universe, because the native linear medium field passes a_0 at $z \geq 84$ ($15.6\, a_0$ at $z = 1100$); slip caps its switch at $u_c \leq 0.22$ / $0.16\, a_0$. So option 1 (the collapse-gated tether) is closed for local Lagrangians.
- **Scope.** ρ-dependent extensions, new fields and a non-variational latch are not closed; a medium held up by dispersion would be a different theory. Lane class: NO CURE, no amendments.

4.3 Lemma L-G and theorem T-O2: cold energy must be the phantom (CFG946)

- **Lemma L-G.** Let ϕ be a field whose gradient is a universal force, with a static equation that is local, second order and translation and rotation invariant. Assume exact uniform-gradient superposition: if ϕ solves it, so does $\phi + \mathbf{b} \cdot \mathbf{x}$. (This is the statement that a satellite in a uniform host field keeps its internal dynamics.) Then the equation depends on ϕ only through $\partial\partial\phi$, and the flat-field amplitude obeys $A(\mu^3 M, \mu R) = \mu^2 A(M, R)$. The deep law needs $\mu^{3/2}$. So no such sector supplies the deep law, and every sector that does supply it has an external-field effect. A sector that supplies part of the force has a share that falls as $\mu^{1/2}$. Deep AQUAL and QUMOND fail superposition, so both have an EFE; a cubic Galileon passes it but has no flat regime.
- **Theorem T-O2: assumptions.** A1: cold energy is pressureless, universally coupled dust in free fall, with Poisson self-gravity (option 2). A2: every other sector is local, second order and translation and rotation invariant, and every force is universal. A3: isolated rotators obey the law with the profile-independent deep amplitude $\sqrt{GM_b a_0}$ for any assembly history, selected by static equations. A4: a uniform external field leaves a satellite's internal dynamics unchanged.
- **Claim.** No construction satisfies A1–A4. By L-G the deep extra field of isolated rotators must be the dust's own field, so cold energy is the law's phantom in the deep regime, contrary to option 2. If one then accepts dust as the phantom, free-falling dust has no selection (half the dust is another equilibrium, missing the law by 0.414 / 0.421), and held dust has its self-gravity cancelled by the holding force (D-T′, which generalises D-T).
- **What it rules out.** All eight constructions are NOT VIABLE. QUMOND-type fields double count the dust and carry an external-field effect of up to -0.58 dex on the test dwarfs. A tether with $c_2 \neq 1$ turns the law into $g_N = R(u) - (1 - c_2)u$, with an acceleration floor. A Galileon has no deep regime. Dispersion-held dust has no selection and reduces to option 3. Plainly, option-2 cold energy is a cold collisionless species in every gravitational respect.
- **Scope.** Four routes lie outside the theorem: non-static, non-local supply (option 3); non-local or higher-derivative field sectors; modified inertia; and charges that violate the equivalence principle. Assembling NGC 3198's phantom from cosmic-mean dust needs a transport length of 25.6 / 26.6 times its radius. Amendment A1 is code and reporting only; one post-result fix corrected a printed fail list, and no class changed.

4.4 Theorem S: label-free tethers hover (CFG945)

- **Hypotheses.** The force on a cold medium element is $\mathbf{f} = -\nabla\Phi + \mathbf{F}(\mathbf{x}, \mathbf{v}; \sigma)$. $\mathbf{F}$ may be any functional of fields sourced by the whole system. It depends on the element only through its position, weakly on its velocity, and on gate scalars σ of its own flow that take their settled value for a stationary medium. There is no membership label.
- **Claim.** The host's medium at $\mathbf{x}$ is cold, static and in equilibrium under the tether force. So any stationary medium element there, a satellite's included, feels zero host force at leading order, while the stars feel the host's pull: $\Delta a = g_{\text{host}} + O(v/c_*) + \delta_{\text{cross}}$. The theorem is exact at leading order in the ratio of satellite field to host

field. The cross terms depend on the model. The escapes are a membership label, support specific to the stream, $O(1)$ velocity dependence, or a non-stationary medium.

- **The collapse gate does not help.** A settled or freely translating satellite medium has $\theta = 0$, so its gate is on: $\Delta a/g_{\rm host} = 1.017$–$1.107$, reproducing CFG941 exactly.
- **What the data allow.** From the record's own satellite statistic (CFG874) under a stripping model, a differential is allowed only up to $\varepsilon_B = 0.344$ (canonical) / 0.317 (alternative). Under the hover, 19 satellites are stripped and the statistic reads -0.355 ± 0.114 (-3.1 standard errors). The room for a satellite-specific deficit is 0.082 dex, against the hover's 0.10–0.73 dex.
- **What survives.** Only a per-label tether (H1) passes: RESOLVED-WITH-COST. Anchoring to the baryons (H2), restoring the equivalence principle (H3) and dispersion support (H4) are NOT RESOLVABLE. A satellite medium abandoned by its stars reaches the gate's ghost band within 1.4–13.5 Myr, and the composite violates the equivalence principle by $\eta = 0.76$–1.04.
- **Disclosure.** A1 is post hoc: the frozen mutation tooth M5 was toothless under CFG874's matched-$g_{\rm bar}$ regression, so a tooth M5b was added before any main number. The lane's mutation run exits 3, as A1 declares.

4.5 Non-local parents of the engine (CFG947)

- **The construction.** Geodesic baryons, untethered cold dust and a conserved non-local sector $\mathcal{S}$ with retarded kernels. Retarded non-local equations come from no action principle. The time-symmetric action form is acausal (a precursor of half the static value before the source switches on), so option 3 cannot meet the one-action rule and causality together.
- **T-M.** Constant weights conserve momentum (identity to 7×10^{-18}). The engine's relocation of its source gives a non-zero term, $\pi^2\varepsilon/4$ of the domain's weight scale (0.25 at dipole fraction $\varepsilon = 0.1$). So the engine is not conservable in asymmetric catchments.
- **T-R.** The engine resolves a system only if $M_b \geq 4.815 \times 10^{10}\, M_\odot$. That switches the law off for 128 of 175 SPARC galaxies; NGC 3198 sits 2% below the threshold. An engine-verbatim parent fits SPARC at 0.257 / 0.262 dex rms, against 0.1003 / 0.0991 for the law.
- **T-X.** Reading the engine's code shows that its phantom is the pointwise QUMOND phantom of the global retained-baryon field, not an own-baryon rule. It therefore carries the satellite external-field effect, -0.10 to -0.73 dex on the test dwarfs at 50–250 kpc, which an earlier lane (CFG478) excludes.
- **Positive, derived.** Zero-sum is forced by conservation (removing it creates $53.65\, M_b$ of $\mathcal{S}$ at switch-on). There is no hover, $\Delta a/g_{\rm host} = 0$ at monopole order, because a parent that changes the source of gravity escapes D-T. Growth is exactly a^1. Slip is $\leq 1.63 \times 10^{-5}$. The law must end at an edge: its enclosed mass at turnaround exceeds the turnaround mass by 1.85–9.72 for $M_b = 10^7$–$10^{15}\, M_\odot$. A gate with no domains leaves negative total density, so domains are forced.
- **Class.** All four candidates (labelled domains, engine-verbatim, turnaround basins, domain-free) are NOT VIABLE: NO-GO for exact parenthood. A1 was pre-result: it corrected the T-M falsifier to the relocation term.

5 The labelled-domain construction, tested (CFG950–CFG952)

The theorems leave one door. Theorem S allows a tether with membership labels, and CFG947 leaves open an approximate non-local parent with labelled domains and a merge rule; CFG945 notes that its per-label tether is the engine's "one tether per catchment" written as fields.

- **As first stated (CFG945 H1, CFG947 N1).** One potential χ_q and one multiplier λ_q per collapsed label q, with action $\sum_q W(|D\chi_q|)$ plus the constraints, and a dipole-free source draw (its centroid tied to the phantom's at all times). The labelled medium falls with its stars ($\Delta a = 0$; tidal residual 1.6×10^{-5} to 2.0×10^{-3}), and Newtonian gravity between distinct collapsed objects is restored, although D-T still cancels it within one label. Costs: no local seeding rule was found; energy and momentum were conserved only between label changes; galaxies must be single-label (two equal labels miss NGC 3198 by 4.12 / 4.61, debris must stay below 3.5×10^{-7}, and a split galaxy errs by 8.6% on NGC 3198 at $f = 0.01$); a gated version inherits R3θ's ghost.

5.1 The construction (CFG950): NOT CONSISTENT AS SPECIFIED

- **Action.** Between label events the medium is split into populations q, with $q = 0$ unlabelled dust:

$$S = \int \sqrt{-g}\Big[\tfrac{c^4 R}{16\pi G} - \sum_q mc^2 n_q - \sum_{q\geq 1}\big(W_{\rm full}(u_q) + \tfrac{3\gamma}{4K^2}\tau_q^2\big) + \sum_{q\geq 1}\lambda_q\big(D^2_{(q)}\chi_q - Kmn_q\big)\Big] + S_b.$$

  There is no θ in the action, so CFG944's ghost does not arise. Every other label's medium is a Newtonian source for label q, so D-T cancels only a label's own gravity. The label sector is non-local and has no action: a θ-latch at first collapse ($\theta \leq 0$) with contact inheritance within L_*; a whole-label ball-domain merge rule, forced because concentric labels tie (T-T); a one-way tidal re-latch; debris $\leq 3.5 \times 10^{-7}$.
- **Passes.** The NGC 3198 law to 2.2×10^{-16}, the BTFR limit 1 and a label edge at 454 / 413 kpc; a dust background, no labels in the linear regime and growth $n = 1$ exactly; own-label satellites fall with their stars ($\Delta a/g_{\rm host} \leq 2.0 \times 10^{-3}$, bound in 18/18); slip $\leq 1.54 \times 10^{-5}$; a forced edge with no SPARC point beyond it; satellite EFE 0. The hover (CFG941/945) and the linear-growth flag (CFG938/941) are gone.
- **Energy: CONDITIONAL.** Momentum is exact through every label event (8×10^{-14}), but energy is not conserved by any dynamical field at label events: assembling NGC 3198's label injects 0.17–0.94 of $|E_{\rm grav}|$, and a merger 0.54–0.70. Theorem T-LE: bound cold medium cannot pay the labelling cost ($\Delta > 0$ for $M_b = 10^7$–$10^{15}\,M_\odot$; $\Delta \approx (97\,{\rm km/s})^2$ on NGC 3198). The books close only through R-E, a declared non-dynamical label energy ($|e_L|/c^2 \leq 1.8 \times 10^{-6}$, post-Newtonian residual $\sim 3 \times 10^{-7}$).
- **Class.** The frozen merger-toy metric f3 FAILS, pinned at the innermost equal-mass shell at every resolution; under the post-hoc amendment A1 it passes. Well-posedness is CONDITIONAL on CFG938's premises. So: NOT CONSISTENT AS SPECIFIED under the frozen rule; LDT CONDITIONAL under post-hoc A1. θ-latch labels hold only 47–76% of the engine's catchment mass, and satellites' own-label media, 41–854 times their stars, act as baryons in their hosts' laws (a new prediction, tested next). The CFG678 scorecard is CFG938's (PASS A5, C2; CONDITIONAL A1, A2, B3, C1; FAIL C3, inherited from the law), with B1 (prescribed labels) and D2 (it needs an LDT engine) UNTESTED. K 7/7; MUTATE 6/6 bite.

5.2 Mergers and clusters (CFG951): FAIL

CFG951 froze before CFG950 appeared, so it used its own working statement (CFG945 H1 and CFG947 N1 / M-R, with full Newtonian sourcing between labels). It is a post-hoc hypothesis test (its targets were read before freezing) with zero new constants.

- **Merger memory.** Cluster members keep their own labelled media (no member coalesces at pericentre), and those media source the host's law. On the 318 eRASS1 clusters $A_{\rm OUTER} = 0.556 \pm 0.048$ ($Z = -9.3$) / 0.496 ± 0.043 ($Z = -11.9$): the prediction overshoots the outer shear, by more than the record's earlier merger-memory reading H-MM (0.765 / 0.744, CFG726). Every bracket (King truncation, own-domain coupling, ICM-sourced halos) still fails eRASS1 OUTER. All three merge variants are INCONSISTENT on both footings.
- **Cores, the Bullet, continuity, UDGs.** $A_{\rm MID}$ falls to 1.213 / 1.089 only by overshooting OUTER (MID/OUTER 2.18, against the law's 1.89); the untethered-host reading (CFG949) fits but needs a selector LDT does not have. The Bullet is UNDERDETERMINED for two variants (no measured pericentre) and INCONSISTENT for instant relabelling. Continuity fails for all: the label-induced phantom changes by 1.16–2.34 per dynamical time (0.70–1.63 in the post-hoc A1 variant). The Coma UDGs keep their labels; the ceiling rises by +0.054 in natural log (an upper bound), still not diagnostic.
- **The core problem.** Inter-label sourcing at full Newtonian strength breaks every multi-label system; dropping it would undo CFG945's restored inter-label gravity. K 9/9; every tooth bites.

5.3 An engine for it (CFG952): SMOKE PARTIAL

A particle-mesh change-set was designed and smoke-tested in a separate 64^3 toy; no production run was made. Labels from the engine's catchment machinery persist through infall (purity 1.0), and the own-domain phantom is exactly free of an external-field effect, while the engine's global phantom has an EFE of up to -0.114 dex in the toy. But the momentum-slaved positive source draw is INFEASIBLE in an asymmetric domain (book residual 81; 6.5 under the post-hoc fallback), a capacity limit of the non-local formulation. A one-dynamical-time merge blend cuts the jump at the trigger to 0.025 (instant relabelling: 2.77), but its peak per-step change, 0.94 as frozen and 0.053 post hoc, exceeds the frozen 0.03 line, and zero-sum breaks on failed Newton steps. CFG947's merge rule

never fires in equal-mass mergers. **The engine's effective RMIN:** its $r_{\rm ON}$ node grid makes the effective RMIN at 1024^3 equal to $1.0471\,{\rm Mpc}/h$, so the law acts only on systems with retained baryons $M_{b,{\rm ret}} \geq 2.19\times10^{11}\,M_\odot$, 4.5 times CFG947's continuous figure; a 256^3 validation cannot reproduce this. Shear tests of labelled domains would need this engine and a 1024^3 run (17–18 h per footing at $\lesssim 5000$ labels); its pre-registration is drafted, not frozen.

5.4 What survives

No complete parent survives: labelled domains remove the hover and the growth flag, but fail cluster outskirts through inter-label sourcing and need a non-dynamical label energy. **The class-A engine, stated plainly.** Its phantom is a global QUMOND field with a satellite external-field effect. Its RMIN cut switches the law off for most SPARC galaxies, and the effective cut at 1024^3 is higher still (CFG952). Its source relocation cannot be conserved. No tested local parent contains it (CFG941, CFG944, CFG946), no non-local parent contains it exactly (CFG947), and the labelled-domain construction, the remaining candidate, is not a complete parent (CFG950–CFG952). The shear pass therefore stands as a computational result of the engine; its physical parent is not established.

6 Data confrontations from the same night

- **Faint gas-rich rotators (CFG940): NOT DIAGNOSTIC (DATA) on both footings.** The 25 McQuinn et al. (2022) dwarfs were tested at their measured radii. None reaches $g_{\rm bar} < 10^{-12}\,{\rm m\,s^{-2}}$ (the lowest is $10^{-11.71}$ with total baryons and $10^{-11.96}$ with enclosed disc baryons, DDO 154), the same wall an earlier lane hit with SPARC.
  - As a secondary result, the rotators are not above the law. With total baryons (A) they sit -0.304 ± 0.068 dex below it (-0.340 ± 0.067 alternative); with enclosed exponential discs (B) -0.083 ± 0.064 (-0.121 ± 0.064). The class is BRACKET-DEPENDENT. On the seven galaxies also in SPARC, the single-point method runs 0.18 dex low under A and matches under B, so the A deficit is not established as a law failure.
  - In the cell $10^{-12} \leq g_{\rm bar} < 10^{-11.5}$, dSphs sit above the rotators by $+0.438\pm0.122$ (A) and $+0.318\pm0.108$ dex (B). That is the kinematic pattern of PAPER56, measured where the dSph excess is only about +0.2 dex. All 25 are above Newton.
- **Coma ultra-diffuse galaxies vs the derived ceiling (CFG948): NOT DIAGNOSTIC (CONTROL-LIMITED) on both footings.** By the no-capture criterion a Coma UDG is Newtonian if born in the cluster, or if stripped suddenly past $f_{\rm crit}$. Born in the field and kept, it sits at most on the law. The exact shell engine puts the ceiling at +0.0176 in natural log (0.008 dex).
  - The nine admitted UDGs sit at +0.760 / +0.669 (natural log), all nine above the law. That is +1.63 / $+1.43\sigma$ above the ceiling with systematics, and +2.84 / $+2.49\sigma$ on statistics alone (FRAGILE).
  - Firmer: the Newtonian reading is EXCLUDED at 5.45σ. Under the criterion, the Coma UDGs must have brought their cold energy in from the field, having fallen in gas-poor or been stripped slowly. Field HI-bearing UDGs have median gas fraction 0.81, so sudden stripping would have made them Newtonian. NGC 1052-DF2/DF4 sit at Newton, as derived ($z = +0.40$ / -0.30).
  - The derived dynamics puts the NGC 1052 group below the law but cannot put Coma above it. That would need S4 or H-EX made real and derivable, which CFG939 found it is not. The frozen sign rule reads PREDICTED only technically. The mutation control fails as a power limit set by the systematics, as declared.
- **Clusters as untethered cold energy (CFG949): EXPLAINS at the primary corner, FRAGILE.** Motivated by D-T, the hypothesis H-UT (post hoc, zero fitted constants) treats clusters as never tethered: baryons plus the whole 5.364× catchment supply as native collapse debris, all Newtonian. This contains no a_0, so it is footing-independent. It was scored on 318 eRASS1 clusters with the record's raw KiDS-1000 shear stack.
  - At the primary corner $A_{\rm MID} = 1.011\pm0.063$ and $A_{\rm OUTER} = 1.047\pm0.089$ (the law gives 1.91 / 1.75 and 1.01 / 0.93). MID/OUTER is 0.97, against the law's 1.89. The core excess over the law inside 0.5 Mpc is 2.9×10^{13} / $2.1\times10^{13}\,M_\odot$, against the needed 2.4×10^{13} / 2.0×10^{13}. The outer mass slope is 0.661, against the observed 0.896 ± 0.209 (law 0.466), which is not discriminating on its own.
  - But only 6 of 40 corners EXPLAIN (4 PARTIAL, 30 FAIL). The shape is right wherever the concentration follows the native collapse rule. The amplitude is set by conventions: a measured-gas-only supply is 2–3 times short, and a virial boundary over-predicts by 1.4–2.2 times. 78% of the primary gas budget is a $\beta = 2/3$ extrapolation beyond R_{500}, which is unmeasured. The inner band (ungraded) is over-predicted by 1.55 times. Six of seven frozen checks pass; one is off by 0.69% in ungraded inner bins.
  - Reading: untethered cold energy at the cosmic ratio is ordinary cold collisionless matter on cluster scales.

Where this works, it works without the framework's distinctive physics, which stays in the tethered (galaxy) regime. It also needs a rule that untethers clusters but not galaxies, and CFG939 found none derivable.

7 What this establishes, and what it does not

- **Established, conditionally.** The record's static field equation has a covariant parent with an isotropic dust background, a derived field-aligned stress, linear well-posedness and negligible slip, under five stated conditions. The medium's identity $\nabla \cdot \mathbf{P} = \rho - \bar{\rho}$ is exact through shell crossing. The derived no-capture criterion puts tidal dwarfs and DF2/DF4 at Newton, and settling conserves energy in the conservative dynamics. The law is the exact unique ground state of the canonical S2 tether (CFG935). Labelled domains remove the satellite hover and the linear-growth flag (CFG950).
- **Ruled out, as plainly.** With the medium's excess as the phantom (S2+S3), the law forbids the medium's linear self-gravity and its gravity between tethered components (D-T). No local gate repairs this without a ghost or a vacuum offset 15–25 times the slip bound (G). Separating cold energy from the phantom forces an external-field effect (L-G, T-O2). Every label-free tether hovers satellites (S). No causal non-local theory has the engine as its exact limit (T-M, T-R, T-X). S4 is not derivable from the single medium. No energy-conserving settling that keeps the released energy inside matter brings an infall start onto the law (CFG935). Labelled domains need a non-dynamical label energy (CFG950), fail cluster outskirts through inter-label sourcing (CFG951), and their non-local engine cannot conserve momentum with a positive draw (CFG952).
- **What survives.** No complete parent. The labelled-domain construction is NOT CONSISTENT AS SPECIFIED under its frozen rule, CONDITIONAL only under a post-hoc amendment, and it fails clusters. The post-hoc reading of CFG935, that the settled law needs a dynamical dark-energy partner plus a thermal reservoir, carries no verdict weight.
- **Not established.** A physical parent for the class-A engine, and so a physical parent for the two-footing shear pass, which remains a computational result. Also not established: a mechanism for the pressure-supported excess (PAPER56); the conservative tether only expels surpluses.
- **Not derived.** $\kappa = \frac{1}{2}$, the cold-to-baryon supply ratio, the retained fraction and L_* remain inputs.

Disclosures. Post-hoc amendments: CFG938 A1 (integrator step fix) and A2 (a check added during scoring); CFG939 A2 (control step and report-only runs, after two failed main runs); CFG945 A1 (a toothless frozen tooth replaced; exit 3); CFG946 A1 (code and reporting, partly post-result, no class change); CFG932 A1 and its post-hoc hypothesis H-ELS; CFG935 A2 (post-result for the dark-energy repair), A3–A5 (post-result compute caps) and its post-hoc runs (no verdict weight); CFG940 A1 (a check that failed its own rounding tolerance); CFG948 A1 (control wording); CFG949 A1 (the gas-extrapolation share) and its post-hoc hypothesis H-UT; CFG950 A1 (the merger-toy metric f3 replaced after the first main run; the frozen class is reported alongside); CFG951 A1 (a reported toy variant after the first main run; no class change), and CFG951 as a whole is a post-hoc hypothesis test; CFG952 A2 (a reported fallback stage after the main results; no grade change). Pre-result amendments: CFG935 A0, CFG939 A1, CFG941 A1, CFG947 A1, CFG952 A1; CFG944 has none. CFG935 A1 is an implementation fix made after its first S2 runs failed numerically, before any valid S2 result. Superseded runs are kept: CFG938 run 1, CFG939's first two main runs, CFG945's first mutation run and a killed first main attempt, the first main runs of CFG946 and CFG947, CFG949's first check and main runs (graded numbers unchanged), CFG935's pre-A1 and pre-A2 records, the first main runs of CFG950 and CFG951, and CFG952's killed first mutation run, crashed first main run and two failed post-hoc attempts. CFG935 stopped four of its own runs by hand. CFG944 reworded three output strings after its first full run; no number changed. Not blind: CFG940 and CFG948 saw record numbers before freezing; CFG935 and CFG950 froze with hand expectations stated. Controls: CFG945's frozen tooth M5 was toothless; CFG948's mutation control fails as a power limit; CFG949 passes six of seven frozen checks; CFG939's S4 class is unassigned by its frozen rules; CFG935's density gate misses a collapse inside $r = 1$.

Public repository lanes (frozen criteria, derivations, scripts, outputs) under `campaign_fresh_gravity/`: CFG938, CFG939, CFG941, CFG944, CFG945, CFG946, CFG947, CFG950, CFG951, CFG952 (structural); CFG932, CFG935 and the ledger row CFG932-audit (energy); CFG940, CFG948, CFG949 (data). Companion notes: PAPER55 (2026-10-19) and PAPER56 (2026-10-20). Data: K. B. W. McQuinn et al., ApJ 940, 8 (2022); F. Lelli, S. McGaugh, J. Schombert, AJ 152, 157 (2016) (SPARC); E. Bulbul et al., A&A 685, A106 (2024) (eRASS1); KiDS-1000 shear as stacked in the record's earlier cluster lanes; the HI-bearing ultra-diffuse galaxy table of L. Leisman et al., ApJ 842, 133 (2017); the Coma ultra-diffuse galaxy table of J. Freundlich et al. (2022). Theory: M. Milgrom, ApJ 270, 365 (1983); J. Bekenstein, M. Milgrom, ApJ 286, 7 (1984) (AQUAL); M. Milgrom, MNRAS 403, 886 (2010) (QUMOND); B. Famaey, J. Binney, MNRAS 363, 603 (2005). AI assistance: Anthropic's Claude (analysis lanes, review, drafting).